

Preliminary Study – Master Thesis Project

Title: **Electoral Closeness and Incumbency Advantage in Spain's
Proportional Representation System**

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Problem Statement and Motivation

The concept of electoral closeness is central to political economy and political science, as it determines both the competitiveness of elections and the conditions under which causal effects of incumbency can be identified. Understanding how narrowly won elections influence subsequent political outcomes helps assess the fairness and dynamics of democratic competition.

In plurality systems, closeness is straightforwardly defined as the vote margin between the winner and the runner-up. This facilitates empirical analysis of incumbency advantage, as the winner is determined by a single cutoff point. By comparing candidates who just barely win with those who just barely lose, researchers can credibly estimate the causal effect of holding office on future electoral outcomes.

In proportional representation (PR) systems, however, seats are allocated according to each party's vote share, and within each party, the ordering of candidates on closed lists determines who enters parliament. A candidate's election outcome thus depends jointly on the party's performance and their position on that list. Identifying who is "barely elected" or "barely not elected" therefore requires a nuanced understanding of list structures, vote shares, and seat allocation formulas.

Spain's national legislature is bicameral, composed of the Congress of Deputies (Congreso de los Diputados) and the Senate (Senado). The Congress, which is the lower house, is elected through a closed-list proportional representation system and holds the decisive legislative authority, including the formation of governments. In contrast, the Senate follows a predominantly majoritarian system, with voters

choosing individual candidates rather than party lists. This study focuses exclusively on elections to the Congress of Deputies, where proportional representation and list dynamics are key to understanding electoral closeness and incumbency advantage.

While most incumbency literature has focused on plurality systems (e.g., Ansolabehere & Snyder, 2002; Lee, 2008; Eggers & Spirling, 2014), recent work by Luechinger, Schelker, and Schmid (2024) has developed a rigorous framework to define electoral closeness and apply Regression Discontinuity Design (RDD) in PR systems. This thesis builds directly on that framework and applies it to Spain, a context where empirical evidence on incumbency advantage remains limited.

Core Research Question and Objectives

Main Research Question

How does narrowly winning a parliamentary seat affect a candidate's future electoral prospects in Spain's proportional representation system?

Sub-Questions

1. How can electoral closeness be accurately measured at the candidate level in Spain's closed-list PR system?
2. Does narrowly winning a seat causally increase the probability of running again and/or being elected in the next election cycle (the incumbency advantage)?

3. How does the incumbency advantage differ across parties and electoral districts of varying sizes (district magnitudes)?
4. To what extent does incumbency affect a candidate's future position on party lists?

Objectives

- Construct a candidate-level measure of electoral closeness tailored to Spain's PR system.
- Identify marginal candidates around seat cutoffs across districts and election years based on this measure.
- Apply Regression Discontinuity Design (RDD) to estimate the causal effect of incumbency among these marginal candidates.
- Develop reproducible Python-based workflows for data preparation and analysis.
- Generate new empirical evidence on incumbency effects under proportional representation.

Definition of the Topic

This thesis aims to adapt and apply the methodology developed by Luechinger et al. (2024) to the Spanish context. It will operationalize electoral closeness at the candidate level, allowing the identification of “barely elected” and “barely unelected” candidates within each district and election year.

While plurality systems define closeness through direct vote margins, proportional systems require integrating both party vote shares and candidate rankings. The project therefore constructs a running variable representing each candidate's effective distance from the seat cutoff, taking into account district magnitude, party performance, and the D'Hondt allocation formula.

By applying the framework developed by Luechinger et al. (2024) to Spain—a closed-list, multi-district system—this study extends their approach and tests its generalizability to a large European democracy.

Initial Considerations Regarding Methodology

Research Design

The empirical strategy follows a Regression Discontinuity Design (RDD) that exploits the discontinuity at the seat-allocation cutoff. Candidates immediately above and below this threshold are comparable in all relevant respects except for the election outcome. This discontinuity provides the quasi-random variation required to estimate the causal effect of incumbency.

Running Variable and Identification

The running variable measures each candidate's relative position around the cutoff and is standardized for comparability across districts. Following the D'Hondt allocation, it indicates how many additional votes a candidate would need to gain or

lose a seat. The causal effect of incumbency is identified by comparing candidates just above and just below this cutoff, assuming that other factors change smoothly around it.

Treatment and Outcomes

- Treatment: being elected to parliament.
- Outcomes: (i) reappearance on a party list, (ii) re-election, and (iii) improved list position in the following cycle.

Estimation Strategy

The analysis will rely on graphical exploration and local linear regression techniques, consistent with the approach outlined in Angrist and Pischke (2015, Chapter 20). Bandwidth sensitivity will be checked to confirm the robustness of the estimated discontinuity.

Data Preparation and Linkage

Each observation in the dataset represents an individual candidate participating in a given election, party list, and province. These candidate-level records include both original variables (e.g., name, party, list position, and election outcome) and derived measures such as the vote margin around the decisive seat cutoff.

The data are based on official election results published by Spain's Ministerio del Interior and have been precompiled by Prof. Dr. Lukas Schmid as part of his comparative research on proportional representation systems. The analysis will focus on cleaning, harmonizing, and linking these records across elections using

Python and the LinkTransformer tool. Manual validation will ensure matching accuracy. All processing will comply with the NDA conditions governing data access.

Validity and Limitations

Potential limitations include incomplete information on re-nominations, limited sample sizes in smaller districts, and endogeneity in party-list formation. These will be addressed through robustness checks and the use of alternative model specifications.

Underlying Data

Two preliminary datasets have been provided for this study:

- “Spain_master_2004–2023.dta”: a Stata file containing aggregated election data, including province-level and party-level results, candidate lists, and seat-allocation information.
- “Spain_final_2004–2019_R.csv”: a CSV file including a subset of candidate-level variables and the key variable vote margin, which represents the vote difference around the decisive seat cutoff.

The current datasets cover the period from 2004 to 2023. Depending on final data availability, earlier election years may be integrated into the completed version of the dataset.

The preliminary datasets contain approximately 1,000 observations and 25 variables at the candidate–election level. They serve as the foundation for data cleaning, linkage, and the computation of normalized closeness measures before the RDD analysis.

Technology, software and applications

The empirical analysis will be carried out entirely in Python, which provides a flexible and transparent environment for data processing, statistical analysis, and visualization.

The workflow is organized into modular scripts and Jupyter Notebooks to ensure clarity and reproducibility. Version control is maintained with Git, allowing the main steps (data cleaning, linkage, and estimation) to be tracked and easily replicated.

The main libraries used are:

- pandas and numpy – for importing, cleaning, transforming, and combining datasets efficiently.
- matplotlib and seaborn – for creating descriptive figures and visualizing results, such as distributions, trends, and regression discontinuities.
- statsmodels and scikit-learn – for statistical estimation, including local linear regression and diagnostic checks used in the Regression Discontinuity Design (RDD).
- LinkTransformer – for automated record linkage, helping to match candidates across election years even when names differ slightly in spelling or formatting.

All scripts will be written in Python within a virtual environment (.venv) inside the project folder.

Project Risks and Challenges

The empirical design relies on detailed multi-year election data and the accurate identification of marginal candidates. Several data and methodological challenges could affect the implementation or interpretation of results. The following risks are therefore considered most relevant.

- **Incomplete linkage across elections**

Correctly linking individual candidates across election years is crucial for identifying incumbents. Variations in name spelling, use of middle names, or changes in surname order may lead to mismatches. This could result in undercounting incumbents or misclassifying reappearing candidates, thereby biasing estimates of the incumbency effect.

Mitigation: Combine automated record linkage (LinkTransformer) with manual verification for ambiguous matches. Randomly audit linkage quality for selected parties and provinces to estimate potential error rates and, if necessary, adjust the analysis accordingly.

- **Limited information on candidate re-nomination**

The data may not include all non-elected candidates who were nominated again but placed too low on the list to be visible in official records. This would underestimate reappearance probabilities and distort treatment effects if the probability of being re-nominated differs systematically between winners and

near-losers.

Mitigation: Cross-check available candidate lists for subsequent elections to capture all re-nominations where possible. Sensitivity analyses can be restricted to candidates from parties with consistent list reporting across years.

- **Small effective sample size around the cutoff**

RDD inference depends on sufficient numbers of candidates close to the decisive seat boundary. In smaller districts or years with few marginal cases, statistical power may be limited, leading to wide confidence intervals or unstable estimates.

Mitigation: Pool elections across years while controlling for fixed effects, use optimal bandwidth selection procedures, and transparently report standard errors and power considerations.

- **Measurement error in the running variable**

The running variable (vote margin around the cutoff) depends on precise vote counts and correct identification of the marginal seat. Small errors in vote totals, seat allocation, or list ordering can shift candidates across the threshold, blurring the discontinuity and attenuating treatment effects.

Mitigation: Reconstruct the cutoff systematically from raw vote data and cross-check against official results. Perform robustness checks excluding districts where small rounding differences could change seat assignment.

- **Heterogeneity across districts and parties**

Incumbency effects may vary substantially depending on district magnitude, party size, or ideological context. Pooling all cases together could mask meaningful variation or yield results that are difficult to interpret.

Mitigation: Conduct subgroup analyses by district size and party family. Report both pooled and stratified estimates to assess whether observed effects are consistent or context-specific.

- **Endogeneity of list positions**

In Spain's closed-list system, candidates are placed in ranked order by their party, and those at the top are most likely to be elected. Because list positions reflect party decisions, they may already capture a candidate's perceived quality or popularity. If these party preferences also influence future list placement, part of what appears to be an incumbency advantage may instead result from non-random selection by parties rather than the causal effect of being elected.

Mitigation: Focus the analysis on candidates near the cutoff, whose election outcomes are almost random from the party's perspective. Additionally, compare results controlling for prior list positions to test whether observed advantages persist after accounting for party placement effects.

- **Data completeness for 2023 elections**

As the most recent election data (2023) are not yet finalized, the current dataset may be incomplete or subject to later revision. Any inconsistencies could affect the comparability of results across years.

Mitigation: Perform the main analysis on the validated 2004–2019 sample and replicate it once the full dataset becomes available. Clearly distinguish between preliminary and final results in reporting.

Work Packages and Schedule

The project is structured into six main work packages that build on one another and cover the full process from literature review to final submission. The sequence provides a logical flow but allows flexibility in timing depending on data availability and analysis needs.

1. Literature review and conceptual framework

The first phase establishes the theoretical and methodological foundation. It involves reviewing key literature on incumbency advantage, electoral closeness, and Regression Discontinuity Design (RDD) in proportional systems. The result is a refined conceptual framework and a clearly defined set of research questions.

2. Data inspection and harmonization

The received datasets are explored, validated, and harmonized to ensure consistency of identifiers, formats, and variables. This includes aligning candidate, party, and province codes across files and years to create a clean and well-documented base dataset for further processing.

3. Candidate linkage and variable construction

Candidate records are linked across elections to identify incumbents and construct the normalized running variable (votemargin). Automated matching with LinkTransformer will be complemented by manual review of uncertain cases. The output is an analysis-ready dataset that enables the RDD estimation.

4. RDD estimation and preliminary analysis

Using the linked data, the main empirical analyses will be conducted.

Graphical and parametric RDD estimations will be applied to visualize and quantify incumbency effects around the electoral cutoff. Intermediate results will be reviewed with the supervisors to ensure validity and robustness.

5. Robustness checks and interpretation

Robustness analyses will test the sensitivity of the results to bandwidth choices, model specifications, and district characteristics. The interpretation will place the findings in the broader context of comparative electoral research and highlight implications for understanding incumbency in proportional systems.

6. Writing and final submission

The final stage consists of integrating all analyses, refining figures and tables, and completing the written thesis in line with university standards and NDA requirements. Supervisory feedback will guide the final revisions before submission.

Supervision is provided by Prof. Dr. Lukas Schmid (main supervisor) and Prof. Dr. Oliver Staubli (co-lecturer), who will offer methodological support throughout the empirical and technical components of the project.

Expected Outcomes

The Master's Thesis is expected to deliver:

- A reproducible, Python-based workflow for measuring and estimating incumbency effects in proportional representation systems.

- A harmonized and well-documented dataset of Spanish parliamentary candidates across multiple election years, suitable for replication or future comparative research.
- Empirical evidence on the magnitude and persistence of incumbency advantage in Spain's proportional representation system, contributing to the broader understanding of electoral competitiveness and candidate dynamics in multi-member districts.

Together, these outcomes will provide both methodological and empirical value, supporting ongoing research on electoral behavior and institutional effects under proportional representation

Declaration of Originality

I hereby declare that I have written this preliminary study independently and exclusively with the aids and support of third parties provided by the Lucerne School of Business, in compliance with the Guidelines for the Use of AI and GenAI Tools in Assessments. All sources and literature used have been fully cited.

Statement on AI Use

Generative AI (OpenAI GPT-5) was used for text editing and formatting support. All conceptual content, structure, and analysis were developed by the author.

Lucerne, October 2025

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